

Contact-Model-Aware Selective Passivation: Results

1 Experimental setup (stub)

Planar two-link MuJoCo robot pressing a spherical pad on a horizontal surface: desired normal force $F_n^d = 5$ N, desired tangential sliding speed $v_t^d = 0.05$ m/s, control rate 1 kHz, surface friction $\mu \approx 0.30$. A scripted human wrench $\mathbf{w}_H$ acts at the end-effector. Controller modes C0–C4, the energy-ledger parameters, and the QP rows are those of the PoC implementation document; this stub exists only so the standalone build compiles with resolved references.

2 Results

Fig. 1 summarizes the experimental system and the implemented pipeline. All experiments share the task of Sec. 1¹ ($F_n^d = 5$ N, $v_t^d = 0.05$ m/s), the ledger parameters, and the robust QP rows; only the task-contact predictor, the scripted human wrench, and the controller mode vary between tests. All results in this section come from one coherent dataset generated by the current robust implementation: the ledger rows constrain the robust lower power estimate $p_i^{\text{pred}}(u) - \|\mathbf{w}_i\| \bar{e}_v$ under the configured one-step velocity prediction-error bound $\bar{e}_v = 0.025$ m/s, so every floor guarantee below is conditional on that bound — which is itself monitored: the largest error measured across all runs is 0.0115 m/s (peak utilization 0.46) with zero violations. The comparison modes are C0 (nominal), C1 (whole-port QP), C2 (residual QP), C3 (dual-ledger task-weighted QP), and C4 (dual-ledger *safe-anchor scalar*: the command is restricted to the line from a feasible minimum-norm safety anchor toward the nominal command, under the identical constraint rows). Table 1 collects the summary metrics of every mode–test combination. Across all runs the QP solves in 24–44 μs (median) with a worst 99th percentile of 57 μs , well within the 1 kHz control budget.

2.1 False depletion under nominal sliding (Test A)

Fig. 2 shows Test A (oracle predictor, no human). C0, C2, and C3 are indistinguishable: $\text{RMSE}(F_n) \approx 0.033$ N, $\text{RMSE}(v_t) \approx 0.0025$ m/s, C3 within 0.02% of C0, and the C3 filter modifies the command on only 0.008% of in-contact steps, with both of its ledgers exactly constant (drift $\leq 10^{-9}$ J). The whole-port baseline C1, in contrast, charges the ordinary sliding-friction dissipation (≈ 0.075 W) to its protected ledger: E_W drains and triggers intervention at $t \approx 5.9$ s. Because the whole-port force is dominated by the (noisy, one-step-delayed) raw contact sample, riding that ledger’s floor is not solver-feasible in this plant: 7658 steps fall back to the bounded emergency action, which stalls tangential sliding while holding the normal push — the pad keeps regulating $F_n \approx 5$ N (Fig. 2b) but the sliding task is permanently halted (Fig. 2c). This is the false-passivation failure mode that motivates contact-model subtraction: an accurate task model leaves the nominal task untouched, whereas whole-port accounting destroys the tangential task on friction alone and cannot even ride its own floor cleanly (claims 1–2).

2.2 Model mismatch stays in the residual port (Test B)

With the regularized friction predictor at $\hat{\mu} = 0.20$ (true $\mu \approx 0.30$) and no human, the mismatch power $p_\Delta \approx -(\mu - \hat{\mu})F_n v_t \approx -25$ mW discharges only the residual ledger (Fig. 3): E_H stays exactly at 0.05 J (drift $\leq 10^{-9}$ J) while E_R bleeds from 0.30 J to 0.023 J over the 15-s run. The first ledger-driven intervention consequently moves

¹Adjust this reference to your setup section.

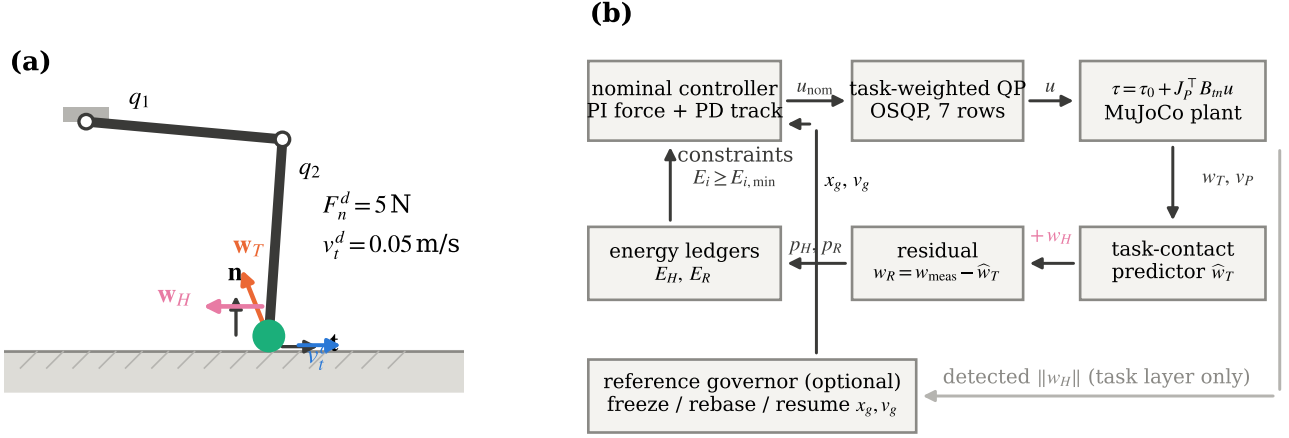


Figure 1: System and signal overview. (a) Planar two-link robot with the spherical contact pad sliding on a horizontal surface; the task-contact wrench $\mathbf{w}_T$ and the scripted human wrench $\mathbf{w}_H$ act at the end-effector, and $(\mathbf{t}, \mathbf{n})$ span the command plane. (b) Implemented pipeline: the task-contact predictor $\hat{\mathbf{w}}_T$ removes model-consistent contact wrench from the measurement, the residual $\mathbf{w}_R$ and human $\mathbf{w}_H$ ports feed the nested energy ledgers, and the task-weighted QP filters the nominal command subject to the robust ledger, human-power, torque, and torque-rate rows. The optional reference governor shapes only the nominal tangential reference and does not enter the certificate.

from 5.95 s (whole-port C1) to 14.59 s (residual, C3): the predictor postpones passivation until *unexplained* contact power has consumed the residual budget, rather than total task friction (claim 3). The attribution identities hold to machine precision throughout ($\mathbf{w}_R = \mathbf{w}_\Delta$ here; $p_R = p_H + p_\Delta$ in every test).

2.3 Human-directed power and energy limiting (Test C)

Fig. 4 is the central quantitative comparison. The nominal controller (C0) works against the 3-N blocking human at $-p_H \approx 0.15 \text{ W}$ for the entire hold and delivers 0.361 J. C2 (residual ledger only) behaves almost identically until its large residual budget is exhausted: the peak human-directed power is unchanged (0.1499 W) and 0.278 J are extracted, because nothing in C2 distinguishes the human from generic unexplained contact. C3 clips the instantaneous power at the $P_H^{\max} = 0.10 \text{ W}$ row (measured peak 0.0997 W), lets the CBF row park E_H just above its floor (run minimum $0.0064 \text{ J} > E_{H,\min} = 0.005 \text{ J}$), and saturates the cumulative output at 0.044 J against the 0.045-J budget — an $8.3\times$ reduction over C0 enforced by the small dedicated ledger rather than by tuning (claim 4). After release the robot recovers nominal sliding in 0.66 s.

2.4 Combined mismatch and human interaction (Test D)

Test D repeats the blocking profile under $\hat{\mu} = 0.20$ (Fig. 5). The human ledger reproduces Test C ($E_H \geq 0.0064 \text{ J}$, peak 0.0997 W), while the residual ledger now pays for both channels and reaches 0.041 J: the stacked decomposition in Fig. 5b shows $-p_R$ separating into the human share $-p_H$ (bounded by the human rows) and the persistent mismatch share $-p_\Delta$. The nested ledgers therefore attribute energy to the correct ports simultaneously. As a diagnostic of the dedicated human ledger’s value: C2 in this same scenario exhausts E_R through the combined drain and logs 1192 solver-infeasible steps around the release, whereas C3 — whose human rows yield first — never lets E_R within 0.02 J of its floor and stays feasible on every step.

2.5 Filter-level task allocation under oblique interaction (Test E1)

Test E1 is a *filter-only allocation ablation*: given the same governor-off nominal controller (the reference governor is disabled for both runs) and the same robust human-energy, residual-energy, human-power, torque, and torque-rate rows, does the full two-dimensional task-weighted QP (C3) preserve the normal-force task better than the safe-anchor scalar restriction (C4)? The oblique human force $\mathbf{w}_H = \frac{3}{\sqrt{2}}(-\mathbf{t} - \mathbf{n}) \text{ N}$ is applied with the standard cosine profile, and E1 is evaluated *only* on the full-magnitude plateau $[t_{\text{start}} + t_{\text{rise}}, t_{\text{start}} + t_{\text{rise}} + t_{\text{hold}}] = [6.5, 8.5] \text{ s}$

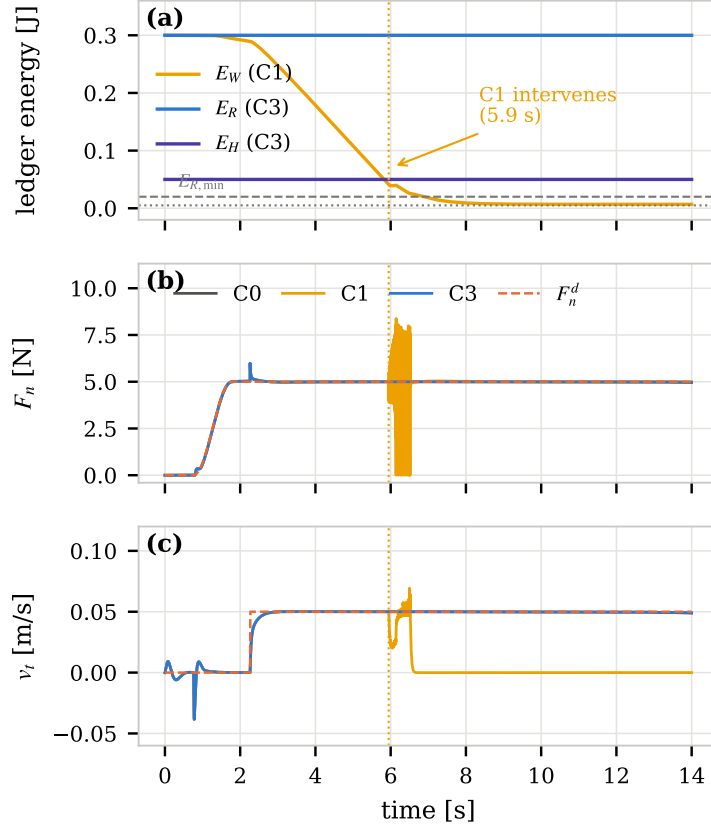


Figure 2: Test A (oracle model, no human): false depletion under nominal sliding. (a) The whole-port ledger E_W of C1 drains on ordinary task friction and triggers intervention at 5.9 s (vertical line), while the C3 residual and human ledgers remain exactly at their initial values. (b) Normal force and (c) tangential velocity: C0 and C3 overlap; C1 stalls the sliding task permanently (via its bounded fallback, which preserves the normal push but marks the run solver-infeasible).

(derived from the scenario configuration): the force-removal transient and the post-release recovery are excluded *by design*, because E1 isolates the geometry of the command filter, not recovery behavior — recovery under a moving reference is governed by the tracking debt the reference accumulates and is treated separately by the reference-governor ablation of Sec. 2.7. Only metrics and plots are clipped to this window; the simulations, logs, and all certificate quantities use the complete 12-s runs.

Both controllers satisfy the identical certificate on the full run (zero infeasible, emergency, prediction-bound, and floor violations; $E_H \geq 0.0064$ J, $E_R \geq 0.256$ J, peak $-p_H = 0.092$ W for both). Within the E1 window their *tangential* behavior is nearly identical — the filtered tangential commands differ by only 0.077 N RMS (0.195 N maximum) against a ~ 23 -N tangential safety action, and $\text{RMSE}(v_t)$ agrees to 2% (0.0457 vs. 0.0465 m/s) — because the same active human-energy boundary largely determines the tangential safety action in both filters. The allocation of the *normal* command is where they part: C3 preserves the normal command more closely (RMS deviation from nominal 0.35 N, vs. 3.91 N for C4; the two filters’ normal commands differ by 2.76 N RMS, 3.41 N maximum), and the normal-force cost follows directly:

$$\text{rmse}_{F_n}^{\text{E1}} = 0.512 \text{ N (C3)} \quad \text{vs.} \quad 2.264 \text{ N (C4)}, \quad (1)$$

a $4.4\times$ difference (C0 reference: 0.112 N). C4’s scalar interpolation from the safety anchor toward nominal cannot trade the two axes independently, so satisfying the human rows drags its normal command away from the force task (Fig. 6c) and F_n sags toward 2.3 N, while C3 spends the entire intervention in the tangential axis (claim 5).

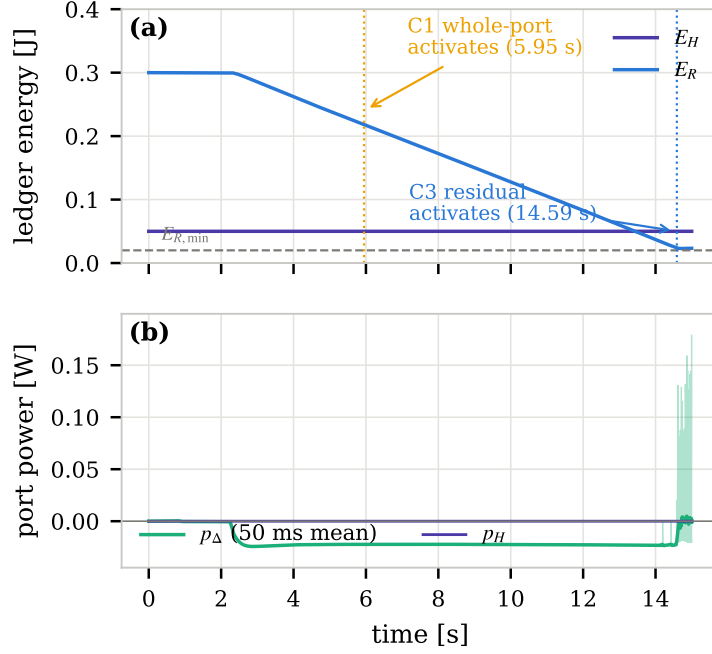


Figure 3: Test B (friction mismatch $\hat{\mu} = 0.20$, no human). (a) The human ledger is exactly constant while the residual ledger depletes; first ledger-driven interventions of C1 (5.95 s) and C3 (14.59 s) are marked. (b) Port powers: the depletion is carried entirely by the mismatch power p_Δ while $p_H \equiv 0$.

2.6 Ledger charging, capping, and discharge (Test F)

Fig. 7 traces the human ledger through a helping (+3 N t) followed by a blocking (−3 N t) pulse. During helping, $p_H > 0$ charges E_H from 0.05 J to the 0.08-J cap, where charging saturates (discharge is never clamped, so the cap is the only nonlinearity). The later blocking pulse first spends the stored credit — the robot resists longer than in Test C — and then approaches the floor (minimum 0.0064 J) with the extraction again power-limited at 0.10 W. Energy received from the human genuinely extends the budget for later interaction, while the cap bounds how much history may accumulate.

2.7 Reference-governor ablation

A time-indexed moving reference keeps advancing while the passivity filter holds the robot, accumulating tracking debt; on release, that debt is repaid as a catch-up burst. Fig. 8 compares C3 with the governor disabled and enabled on the blocking scenario: the burst reaches 0.68 m/s and a 24.2-N normal-force transient without the governor (peaking during the force ramp-out; still 11.6 N after the force has fully vanished), versus 0.050 m/s and 5.1 N with it — the post-release F_n overshoot is reduced by 98.6 % and the velocity stays within 0.2 % of v_t^d . Panel (c) shows the mechanism: the governed reference freezes and continuously rebases onto the end-effector during the interaction (even following the human’s back-push), then resumes with a cosine ramp from the rebased position; both position and velocity references remain continuous. The certificate quantities are unaffected by construction (the governed run keeps $E_H \geq E_{H,min}$ with a peak extraction of only 0.010 W, since yielding removes the incentive to fight the human). The governor is purely a task/recovery policy (claim 6); it is disabled in the certificate tests A–F above so the ledger rows are genuinely exercised, and in any end-to-end behavioral comparison the *same* governor must be applied to C3 and C4 alike — which is why E1 excludes release behavior rather than letting an ungoverned catch-up transient contaminate a filter-level comparison.

2.8 Friction-model sweep

Fig. 9 sweeps the predictor coefficient $\hat{\mu} \in \{0.20, 0.25, 0.30, 0.35\}$ for C3. Residual depletion scales monotonically with the model error: $\hat{\mu} = 0.20$ reaches the floor region ($E_R \rightarrow 0.023$ J), 0.25 consumes roughly half the budget, and $\hat{\mu} \geq 0.30$ produces no mismatch-driven depletion at all — those predictors only *charge* the ledger (the plant’s

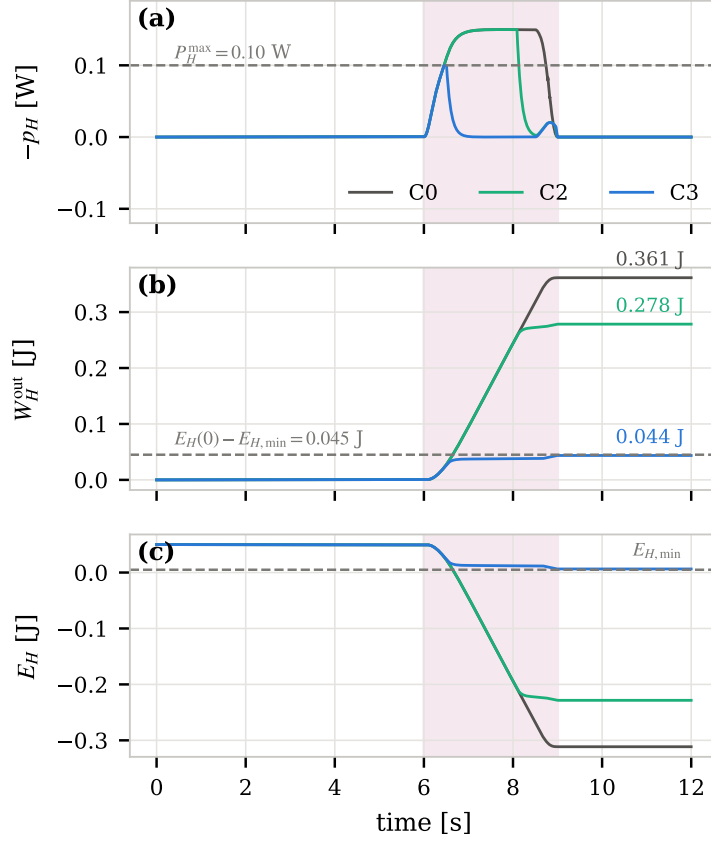


Figure 4: Test C (oracle model, 3-N blocking human, shaded window): human-port limiting across C0/C2/C3. (a) Instantaneous extracted power $-p_H$ with the 0.10-W row; (b) cumulative human-directed energy W_H^{out} against the 0.045-J budget; (c) the human ledger: C3 stays above $E_{H,\min}$, while the (unconstrained, observed) ledgers of C0 and C2 run far below it.

regularized effective friction sits slightly below the nominal $\mu = 0.30$, so their $p_\Delta \geq 0$ while sliding). With a blocking human the first intervention is dominated by the human rows and is therefore independent of $\hat{\mu}$ (6.45 s), while without a human it occurs only for the largest mismatch (14.59 s at $\hat{\mu} = 0.20$). With the human present, task tracking is exactly flat across the sweep; without the human it is flat except at $\hat{\mu} = 0.20$, whose elevated RMSE stems entirely from the brief post-intervention stall after 14.59 s rather than from degraded sliding before it — moderate model error costs residual budget, not task performance. The human-ledger floor is respected at every $\hat{\mu}$.

2.9 Summary

Table 1 condenses the comparison on the current robust dataset. Whole-port accounting (C1) falsely consumes its budget on predictable friction and permanently stalls the task it was meant to protect (its floor cannot even be ridden solver-feasibly in this plant — 7658 and 8658 bounded-fallback steps in Tests A and B, reported rather than hidden); contact-model subtraction (C2/C3) postpones intervention until the interaction is unexplained by the task model; the dedicated human ledger and power row (C3) reduce peak and cumulative human-directed output by factors of 1.5 and 8.3 relative to C0 while the residual ledger absorbs model error (and keeps C3 feasible in Test D, where C2 fails); the full task-weighted QP preserves normal-force regulation 4.4× better than the safe-anchor scalar over the E1 plateau while producing an almost identical tangential safety action; and the reference governor removes the post-release burst without touching any certificate quantity. A 32-s stress run with six alternating helping/blocking pulses sustains all of the above with zero infeasible steps, zero bound or floor violations, and bit-exact ledger accounting throughout.

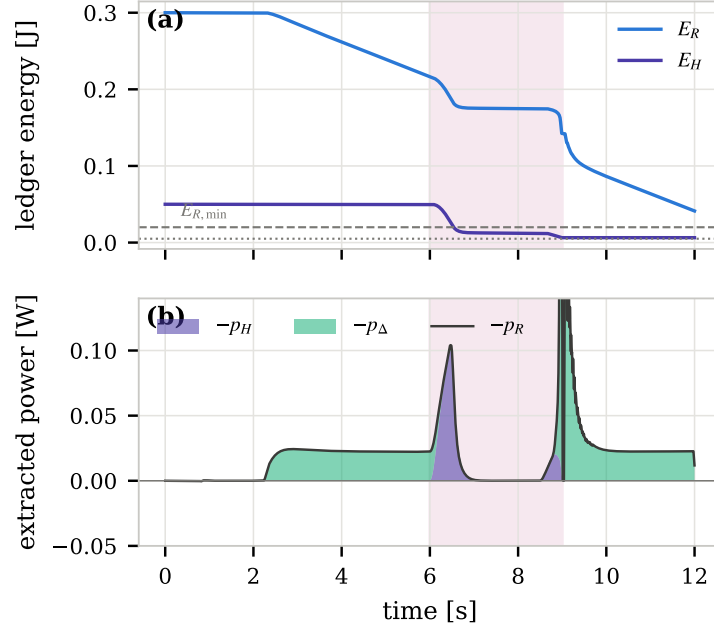


Figure 5: Test D (mismatch $\hat{\mu} = 0.20$ plus blocking human). (a) Both ledgers respond: E_H tracks only human power, E_R tracks human plus mismatch. (b) Extracted-power decomposition $-p_R = -p_H - p_\Delta$ (50-ms means; stacked areas sum to the black $-p_R$ trace).

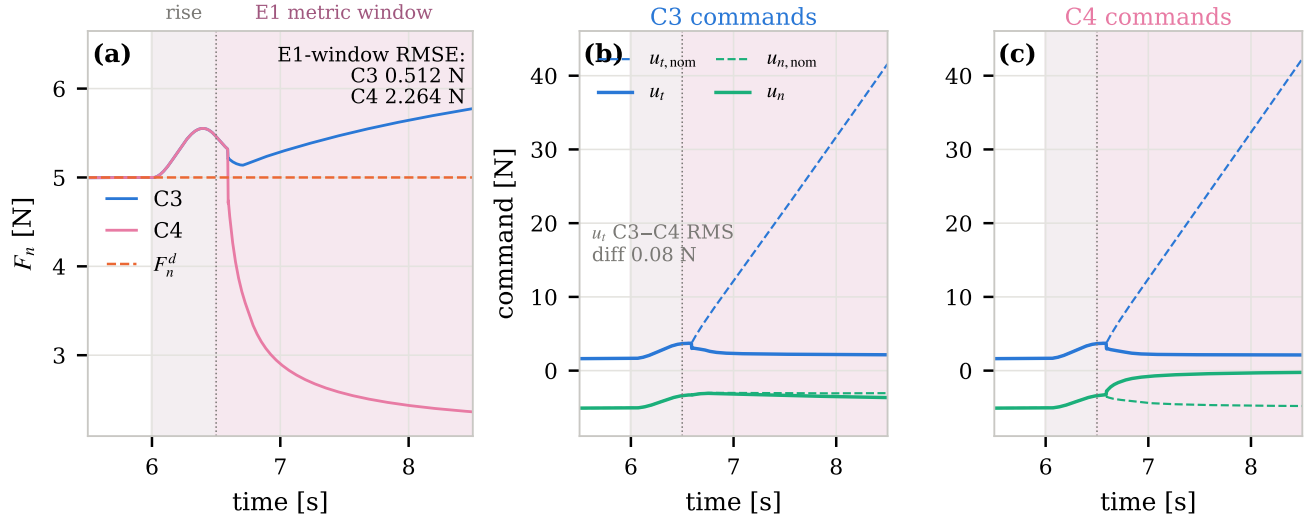


Figure 6: Test E1, evaluated over the full-magnitude oblique-interaction plateau (6.5–8.5 s, shaded; the lighter band is the force-rise context). Both controllers use the same governor-off nominal reference and the same robust passivity rows; force removal and post-release recovery are excluded by design. (a) Normal-force regulation with the freshly computed E1-window RMSE. (b) C3 commands: the intervention is carried almost entirely by the tangential axis while the normal command stays close to its nominal. (c) C4 commands: the safe-anchor scalar moves both axes along one line, pulling the normal command off the force task. The full-dimensional C3 QP produces nearly the same tangential safety action as C4 (0.08 N RMS difference) while preserving the normal-force task more closely than the safe-anchor scalar restriction.

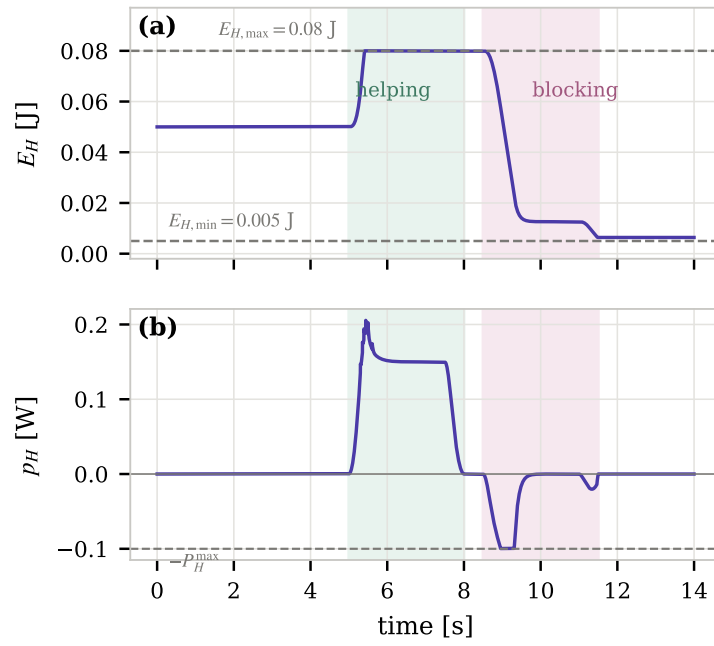


Figure 7: Test F (helping then blocking; green/pink windows). (a) E_H charges to the 0.08-J cap, plateaus, and later discharges toward the floor. (b) The human-port power showing the charge ($p_H > 0$) and bounded discharge ($p_H \geq -0.10$ W) phases.

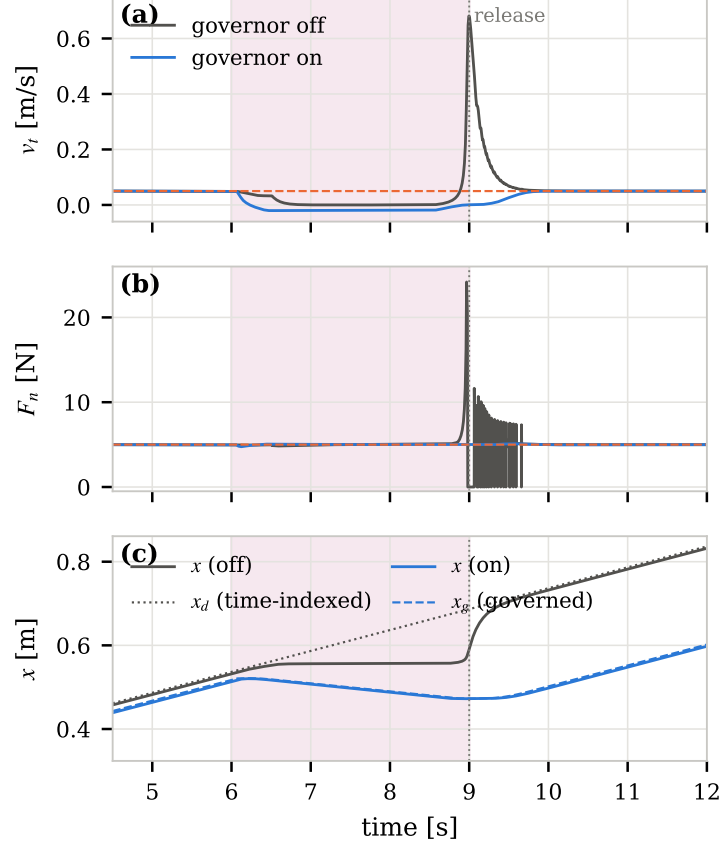


Figure 8: Reference-governor ablation (blocking scenario, C3; release marked at 9 s). (a) Tangential velocity: the 0.68-m/s catch-up burst is replaced by a ramp bounded by v_t^d . (b) Normal force: the release transient is suppressed from a 24.2-N peak (reached during the force ramp-out) to 5.1 N. (c) References: the governed x_g freezes and rebases onto the end-effector during the interaction instead of tracking the time-indexed x_d .

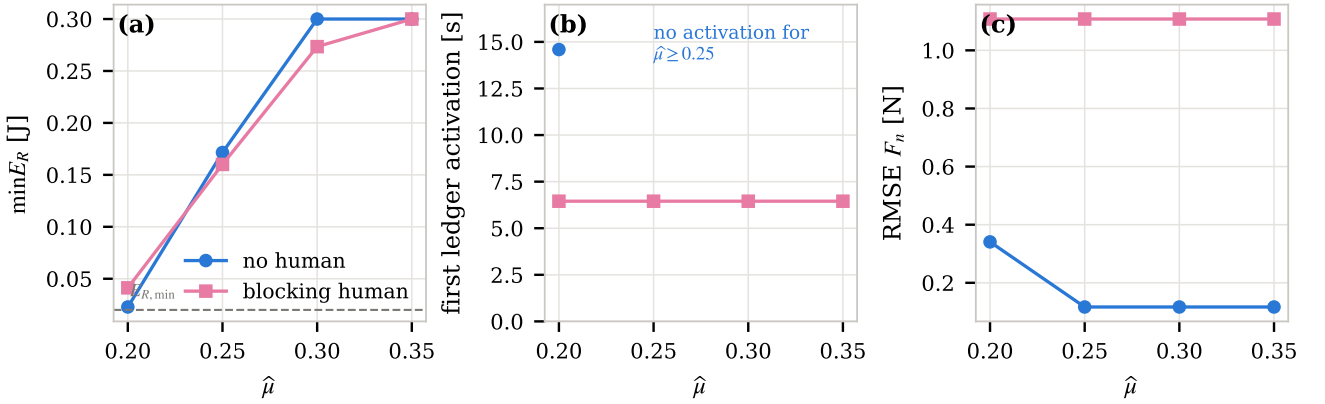


Figure 9: Friction-predictor sweep for C3 with and without the blocking human. (a) Minimum residual-ledger level: mismatch-driven depletion scales with $|\mu - \hat{\mu}|$ and vanishes for $\hat{\mu} \geq 0.30$ (the blocking-human family still discharges through the human share of p_R). (b) First ledger-driven intervention: human-driven (6.45 s) when present, mismatch-driven (14.59 s at $\hat{\mu} = 0.20$) or absent otherwise. (c) Sliding-phase RMSE(F_n): flat with the human at every $\hat{\mu}$; the no-human point at $\hat{\mu} = 0.20$ is elevated only by the brief post-intervention stall.

Table 1: Summary metrics for all controller modes and tests (current robust implementation). RMSE values are over the sliding phase, EXCEPT the Test E1 rows (marked $\dagger$), which report $\text{rmse}_{F_n}^{\text{E1}}$ and $\text{rmse}_{v_t}^{\text{E1}}$ over the full-force plateau $[t_{\text{start}} + t_{\text{rise}}, t_{\text{start}} + t_{\text{rise}} + t_{\text{hold}}]$ only, because E1 is a filter-allocation ablation that excludes the release transient by design; all other columns (peak power, energies, counts) always use the complete unclipped run. $-p_H^{\text{max}}$ is the peak instantaneous power delivered to the human port; W_H^{out} is the maximum of the SIGNED cumulative human-directed output energy (net; in Test F the helping phase offsets the later extraction, so it stays at zero); $\min E_H$, $\min E_R$ are raw ledger minima (discharge is never clamped); “act.” is the fraction of in-contact steps on which the filter modified the command; t_{act} is the first ledger-driven intervention; “fail” counts QP solve failures handled by the bounded emergency fallback. Test S is the 32-s alternating helping/blocking certificate stress run (reference governor enabled).

Test	Mode	RMSE F_n [N]	RMSE v_t [m/s]	$-p_H^{\text{max}}$ [W]	W_H^{out} [J]	$\min E_H$ [J]	$\min E_R$ [J]	act. [%]	t_{act} [s]	fail
A	C0	0.033	0.0025	0.000	0.000	0.050	0.300	0.00	—	0
A	C1	0.424	0.0400	0.000	0.000	0.050	0.300	58.94	5.95	7658
A	C2	0.033	0.0025	0.000	0.000	0.050	0.300	0.01	—	0
A	C3	0.033	0.0025	0.000	0.000	0.050	0.300	0.01	—	0
A	C4	0.033	0.0025	0.000	0.000	0.050	0.300	0.01	—	0
B	C0	0.117	0.0025	0.000	0.000	0.050	0.014	0.00	—	0
B	C1	0.407	0.0409	0.000	0.000	0.050	0.218	61.82	5.95	8658
B	C2	0.341	0.0030	0.000	0.000	0.050	0.023	1.13	14.59	0
B	C3	0.341	0.0030	0.000	0.000	0.050	0.023	1.13	14.59	0
C	C0	0.219	0.0051	0.150	0.361	-0.311	-0.061	0.00	—	0
C	C2	0.543	0.0242	0.150	0.278	-0.228	0.022	7.63	8.09	0
C	C3	1.108	0.0694	0.100	0.044	0.006	0.256	22.40	6.45	0
C	C4	2.215	0.0690	0.100	0.044	0.006	0.256	22.42	6.45	0
D	C0	0.219	0.0051	0.150	0.361	-0.311	-0.276	0.00	—	0
D	C2	1.421	0.0445	0.150	0.189	-0.139	0.019	28.75	7.37	1192
D	C3	1.108	0.0694	0.100	0.044	0.006	0.041	22.40	6.45	0
D	C4	2.215	0.0690	0.100	0.044	0.006	0.049	22.42	6.45	0
E †	C0	0.112	0.0021	0.107	0.259	-0.209	0.041	0.00	—	0
E †	C3	0.512	0.0457	0.092	0.044	0.006	0.256	21.12	6.59	0
E †	C4	2.264	0.0465	0.092	0.044	0.006	0.256	21.11	6.59	0
F	C0	0.286	0.0061	0.150	0.000	-0.281	0.139	0.00	—	0
F	C3	0.938	0.0571	0.100	0.000	0.006	0.300	19.01	8.95	0
S	C3	0.044	0.0139	0.010	0.000	0.050	0.300	0.00	—	0